FDS EXPERIMENT 19

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Experiment to understand Logistic Regression for the given data set.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
```

```
data = {
    "Study_Hours": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "Result": [
        "Fail", "Fail", "Fail", "Fail", "Fail",
        "Pass", "Pass", "Pass", "Pass", "Pass"
    ]
}

df = pd.DataFrame(data)

print(df)
```

```
   Study_Hours Result
0            1   Fail
1            2   Fail
2            3   Fail
3            4   Fail
4            5   Fail
5            6   Pass
6            7   Pass
7            8   Pass
8            9   Pass
9           10   Pass
```

```
X = df[["Study_Hours"]]
y = df["Result"]
```

```
model = LogisticRegression()

model.fit(X, y)

print("Model trained successfully!")
```

```
Model trained successfully!
```

```
new_student = pd.DataFrame({
    "Study_Hours": [7]
})

prediction = model.predict(new_student)
probability = model.predict_proba(new_student)

print("Predicted Result:", prediction[0])
print("Probability of Fail:", round(probability[0][0], 3))
print("Probability of Pass:", round(probability[0][1], 3))
```

```
Predicted Result: Pass
Probability of Fail: 0.144
Probability of Pass: 0.856
```